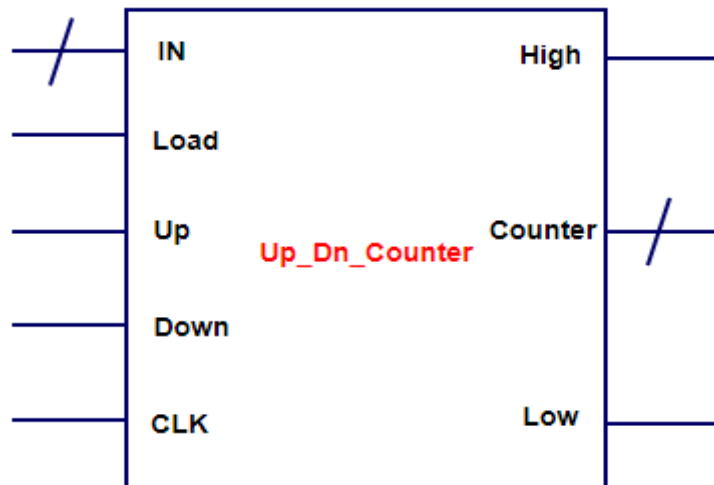


Assignment 2

Specification:

- 5-bit counter
- Count Value loaded from "IN" port on a positive clock edge when "Load" signal is high, Load has highest priority
- Count value incremented by 1 on a positive clock edge when "Up" signal is high.
- Count value decremented by 1 on a positive clock edge when "Down" signal is high, Down has higher priority than "Up" signal
- Increment stops at 31
- Decrement stops at 0
- "High" flag active high whenever count value is 31
- "Low" flag active high whenever count value is 0



Required

- Write a Verilog code equivalent to the above specifications
- Draw the synthesis diagram of your code using Draw.io Tool
- Write a simple Verilog testbench to test your design code.